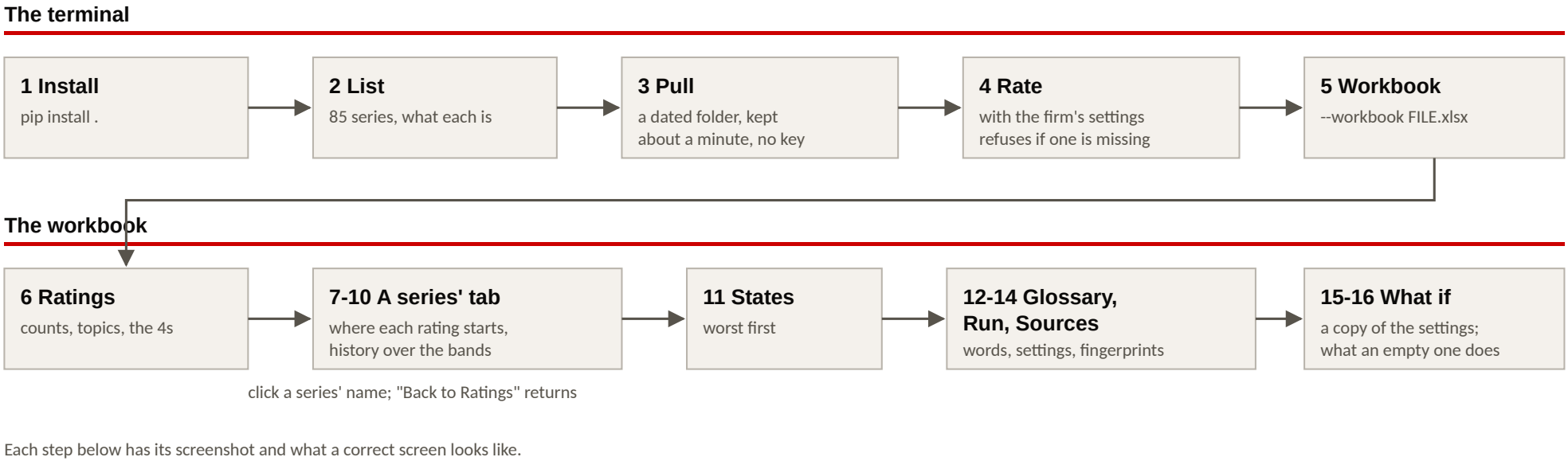
Refresh the economic conditions workbook

The monthly job: download the public data, rate it with the firm's settings, write the workbook and read it for the narrative. Walked end to end on 16 October 2026, twice: once to find what was wrong, and once after the fixes, which is the run pictured here. About five minutes, most of it the download.

You need: this repository checked out, Python 3.10 or later, and internet access to FRED, the Bureau of Labor Statistics, the Federal Reserve Board and FHFA. No key or password. A spreadsheet program: the pictures are LibreOffice; Excel shows the same workbook.

The rule that does not bend: only public data goes in. Nothing from the bank's own books.



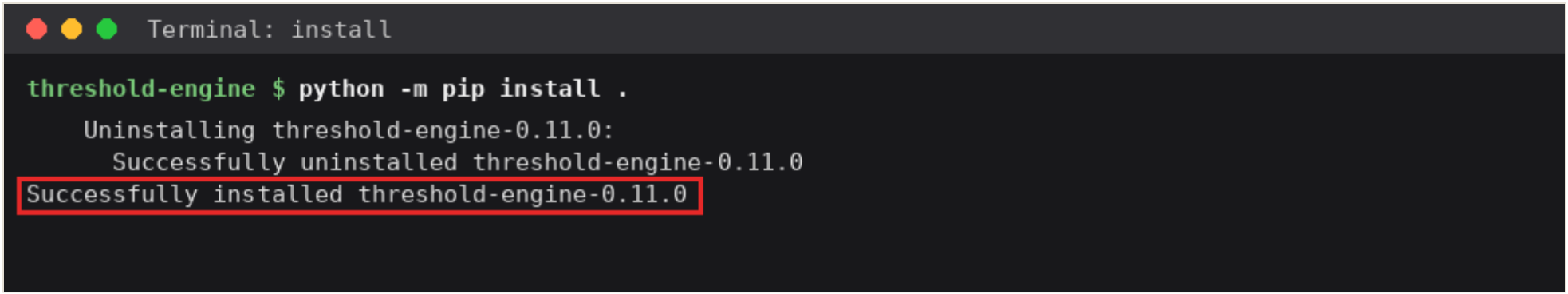
Part 1: the terminal

1. Install the tool

In the `threshold-engine` folder, run:

```
python -m pip install .
```

A correct screen ends `Successfully installed threshold-engine-0.11.0` (the number may be higher).

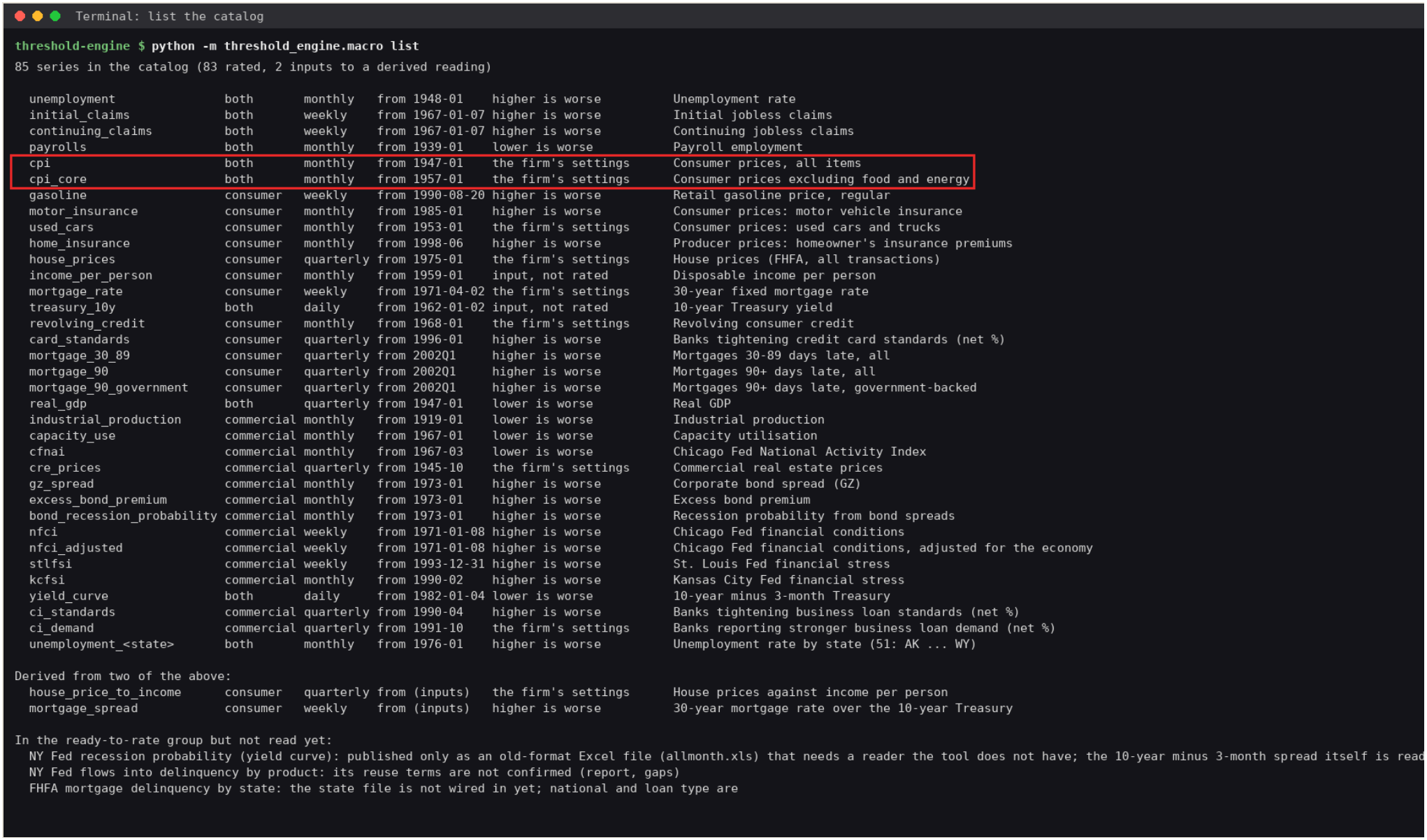


The install ends "Successfully installed"

2. See what it reads

```
python -m threshold_engine.macro list
```

A correct screen lists 85 series, then two readings made from two series, then three it does not read yet and why. Nine series say *the firm's settings* where the direction of stress would be: the firm chose those on 16 October 2026, in `docs/macro/settings.json`. Two say *input, not rated*.

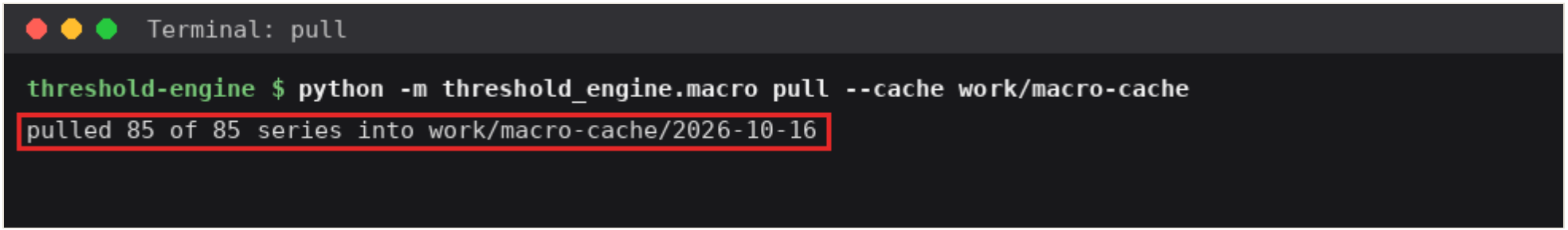


The catalog; the nine directions the firm set are ringed

3. Download the data

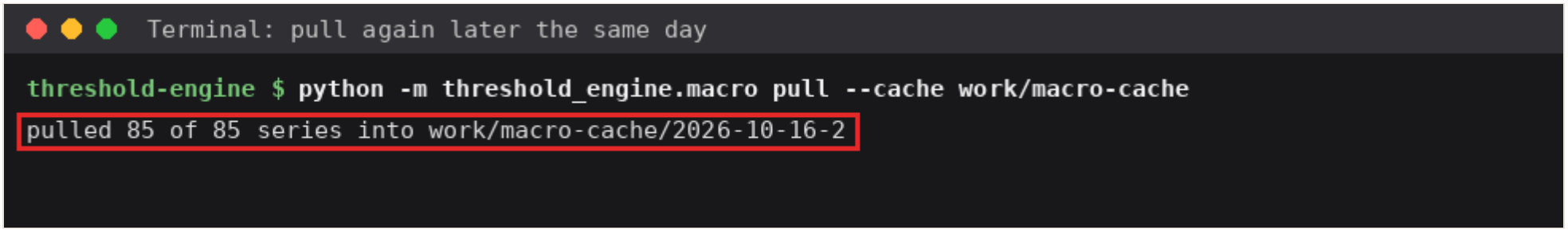
```
python -m threshold_engine.macro pull --cache work/macro-cache
```

It takes about a minute. **A correct screen** says `pulled 85 of 85 series into work/macro-cache/` and today's date. If a series fails, it says `FAILED`, the series and why, pulls the rest, and the command ends with an error. Rate anyway and the failed series is listed as not rated.



85 of 85 series pulled into today's folder

Every download is kept in its own folder. Pull again the same day, after a release, and the new one goes to today's date with `-2` after it; the earlier folder is untouched, so a workbook made from it can still be traced to what it read.

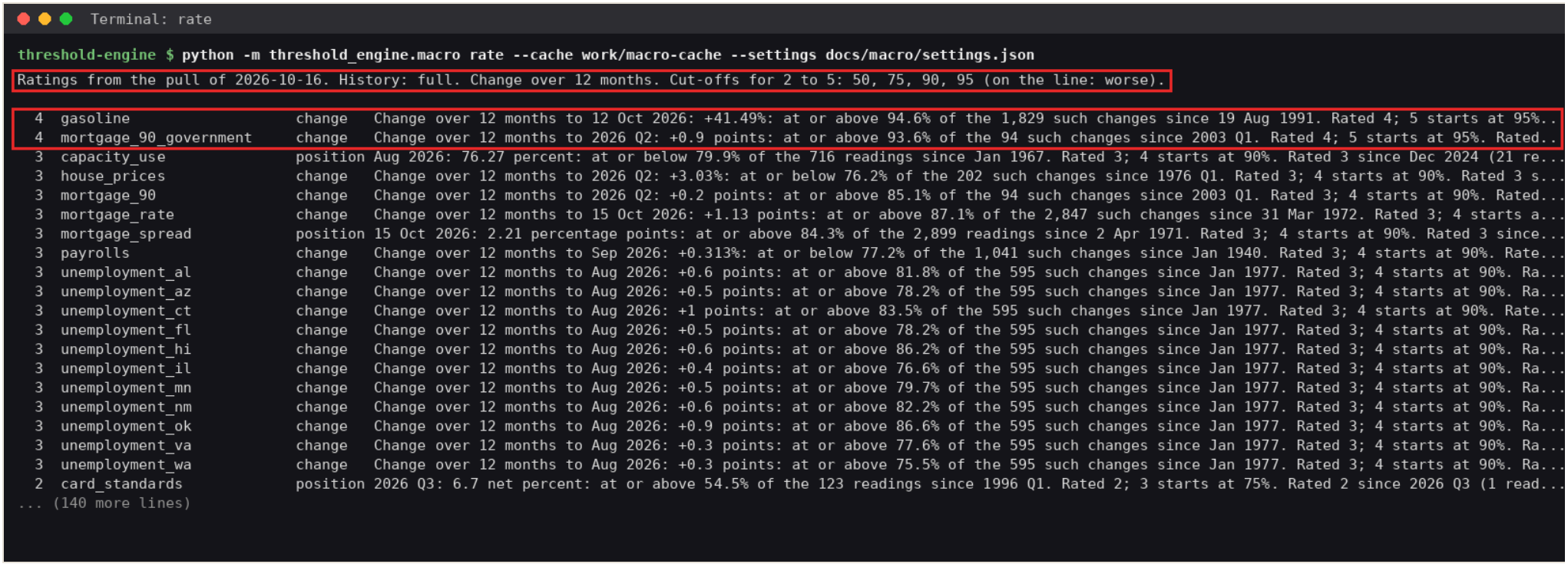


A second pull the same day goes to its own folder

4. Rate

```
python -m threshold_engine.macro rate --cache work/macro-cache --settings docs/macro/settings.json
```

A correct screen opens with one line naming the pull, the history, the change horizon and the cut-offs, then every rating, highest first, each with the sentence behind it. The last line is the Sahm reading, shown and never rated.



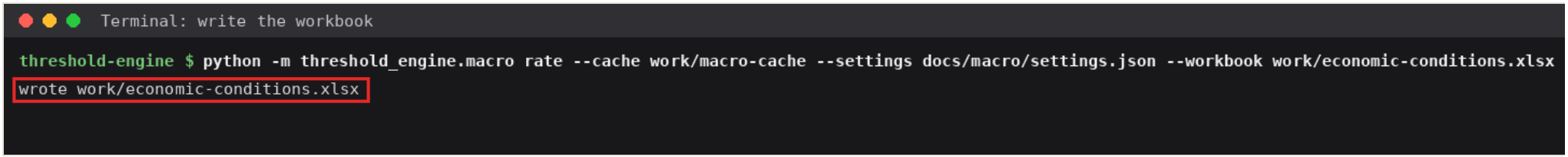
The ratings, highest first; the first line names the settings

5. Write the workbook

Run the same command with `--workbook` and a file name:

```
python -m threshold_engine.macro rate --cache work/macro-cache --settings docs/macro/settings.json --workbook work/economic-conditi
ons.xlsx
```

A correct screen ends `wrote work/economic-conditions.xlsx` after about ten seconds.



The workbook written

Part 2: the workbook

6. Read the Ratings tab

Open the workbook; it opens on **Ratings**. At the top, the count of national readings at each rating: on 16 October 2026, none at 5 High, two at 4, six at 3. Below, every series under its topic (jobs; prices and insurance; housing and household credit; business and the economy; markets and financial conditions), the highest-rated first, with its latest value, both ratings and the sentence that stands out.

Check: the counts add up to the number of national readings (56). Find each 4 and 5 under its topic.

Economic conditions

Public data, pulled 2026-10-16. Each series against its own history: 1 Low to 5 High.

Readings at each rating, national series	5 High	4 Mod-High	3 Moderate	2 Mod-Low	1 Low
	0	2	6	17	31

Sahm reading — 0.00 in Sep 2026. Sahm's published trigger is 0.50; shown, not rated.

Level: where the latest reading sits in the series' own history. Change: how its change over 12 months compares with every such change in that history. A rating of 2 starts at the 50th percentile, 3 at the 75th, 4 at the 90th and 5 at the 95th. Click a series

Jobs

Series	Latest	Value Unit	Level	Level percentile	Change over 12 months	Change	Change percentile	History from	What st
Payroll employment	2026-09-01	159,044 thousands of jobs	change only		+0.31%	3 Moderate	77.2%	1940-01	Change starts at
Unemployment rate	2026-09-01	4.2 percent	1 Low	23.3%	-0.2	2 Moderate-Low	56.5%	1948-01	Change starts at
Continuing jobless claims	2026-10-03	1,710,000 claims	1 Low	11.4%	-11.31%	1 Low	26.2%	1967-01	Change starts at
Initial jobless claims	2026-10-10	199,000 claims	1 Low	2.9%	-10.36%	1 Low	26.6%	1967-01	Change starts at

Prices and insurance

Series	Latest	Value Unit	Level	Level percentile	Change over 12 months	Change	Change percentile	History from	What st
Retail gasoline price, regular	2026-10-12	4.33 dollars a gallon	change only		+41.49%	4 Moderate-High	94.6%	1991-08	Change 4; 5 sta
Consumer prices, all items	2026-09-01	336.3 index 1982-84=100	change only		+3.72%	2 Moderate-Low	66.9%	1948-01	Change at 75%.
Consumer prices excluding food and energy	2026-09-01	338.9 index 1982-84=100	change only		+2.57%	1 Low	45.4%	1958-01	Change at 50%.
Consumer prices: motor vehicle insurance	2026-09-01	850.39 index 1982-84=100	change only		-4.98%	1 Low	1.6%	1986-01	Change 50%. Ra
Consumer prices: used cars and trucks	2026-09-01	182.18 index 1982-84=100	change only		-1.46%	1 Low	30.5%	1954-01	Change at 50%.
Producer prices: homeowner's insurance premiums	2026-09-01	279.51 index Jun 1998=100	change only		+0.62%	1 Low	4.6%	1999-06	Change at 50%.

Housing and household credit

Series	Latest	Value Unit	Level	Level percentile	Change over 12 months	Change	Change percentile	History from	What st
Mortgages 90+ days late, government-backed	2026-04-01	2.8 percent of mortgages	2 Moderate-Low	71.4%	+0.9	4 Moderate-High	93.6%	2002-01	Change starts at

The Ratings tab: the counts, and the two 4s ringed



The counts at each rating, close up



What stands out for gasoline, close up: the sentence behind its higher rating

7. Open a series: gasoline

Click a series' name; it opens its tab. Each tab gives both ratings with what drove them, then **Where each rating starts**, in the series' own units, and how many readings in its history took each rating. Under that, its history drawn over the five rating bands.

Check: the latest value sits between where its rating starts and where the next one starts. Gasoline's change over a year, +41.49%, is at or past 34.43% (where 4 starts) and short of 42.4% (where 5 starts), so it is a 4.

Retail gasoline price, regular

EIA (weekly survey). dollars a gallon, weekly. Pulled 2026-10-16.

Back to Ratings

Level

change only

Change only: its level climbs over the decades with prices and the economy, so its place in its own history says nothing.

Change over 12 months

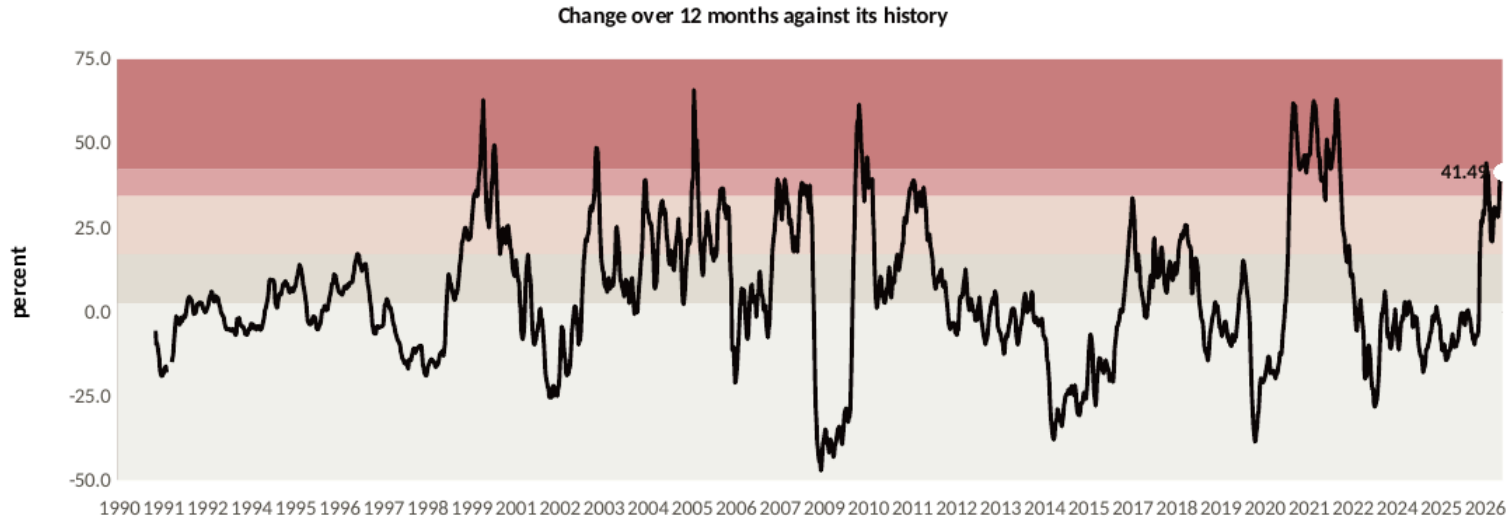
4 Moderate-High **+41.49%** 2026-10-12

Change over 12 months to 12 Oct 2026: +41.49%: at or above 94.6% of the 1,829 such changes since 19 Aug 1991. Rated 4; 5 starts at 95%. Rated 4 since 5 Oct 2026 (2 readings).

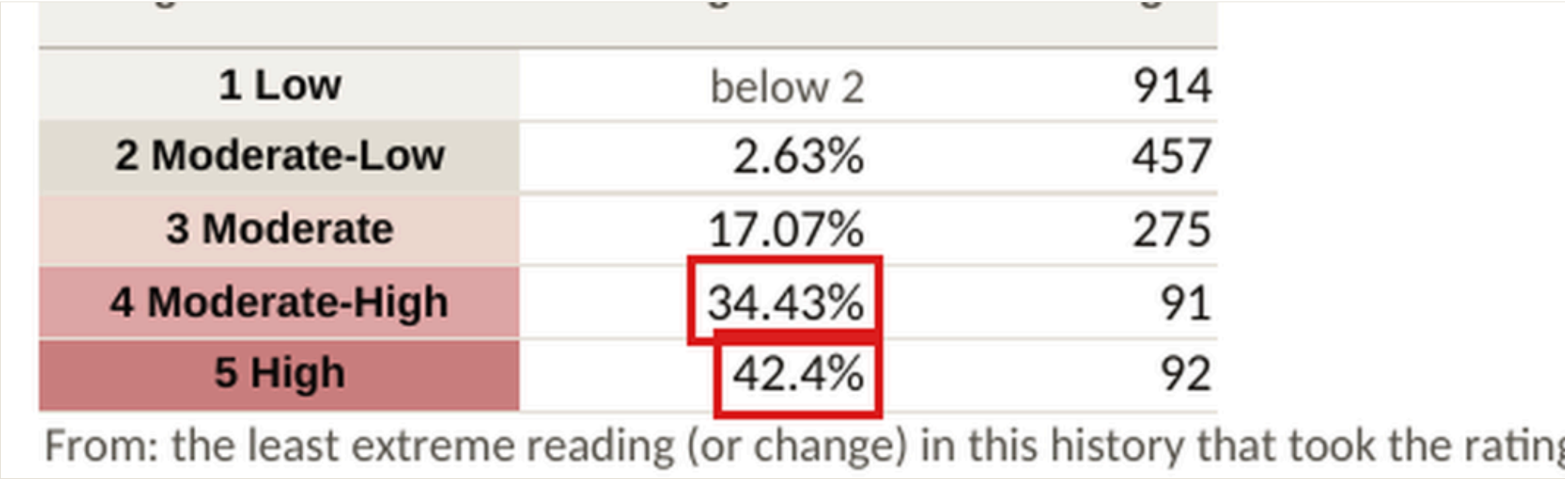
Where each rating starts

Rating	Change from	Changes
1 Low	below 2	914
2 Moderate-Low	2.63%	457
3 Moderate	17.07%	275
4 Moderate-High	34.43%	91
5 High	42.4%	92

From: the least extreme reading (or change) in this history that took the rating; every one further in the direction of stress took that rating or higher.



Gasoline: the rating, the change and where 4 and 5 start, ringed



1 Low	below 2	914
2 Moderate-Low	2.63%	457
3 Moderate	17.07%	275
4 Moderate-High	34.43%	91
5 High	42.4%	92

From: the least extreme reading (or change) in this history that took the ratin

Where each rating starts, close up

8. The other 4: government-backed mortgages 90+ days late

The level, 2.8% of mortgages, is a 2 (2 starts at 2.2, 3 at 3.1). The change over a year, +0.9 points, is a 4 (4 starts at 0.6, 5 at 1.0). A series can be calm on one reading and stretched on the other; the Ratings tab shows both side by side for that reason.

Mortgages 90+ days late, government-backed

FHFA (National Mortgage Database). percent of mortgages, quarterly. Pulled 2026-10-16.

Back to Ratings

Level

2 Moderate-Low **2.8** 2026-04-01

2026 Q2: 2.8 percent of mortgages: at or above 71.4% of the 98 readings since 2002 Q1. Rated 2; 3 starts at 75%. Rated 2 since 2025 Q4 (3 readings).

Change over 12 months

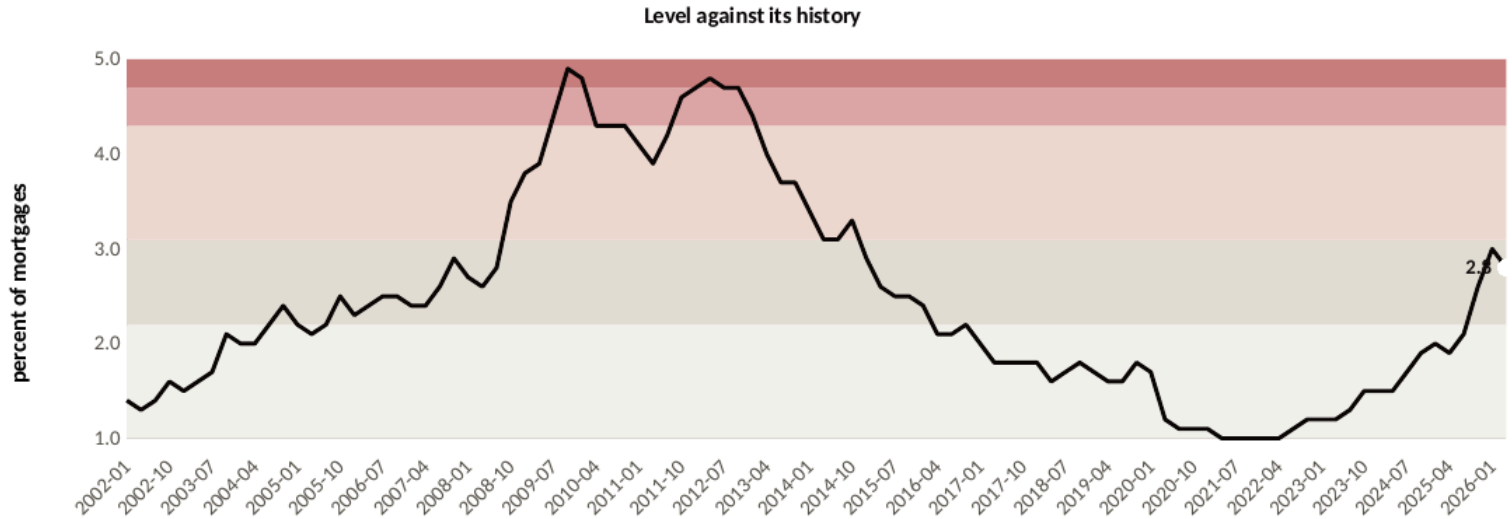
4 Moderate-High **+0.9** 2026-04-01

Change over 12 months to 2026 Q2: +0.9 points: at or above 93.6% of the 94 such changes since 2003 Q1. Rated 4; 5 starts at 95%. Rated 4 since 2026 Q2 (1 reading).

Where each rating starts

Rating	Level from	Readings	Change from	Changes
1 Low	below 2	48	below 2	46
2 Moderate-Low	2.2	25	0.1	15
3 Moderate	3.1	13	0.3	22
4 Moderate-High	4.3	6	0.6	5
5 High	4.7	6	1.0	6

From: the least extreme reading (or change) in this history that took the rating; every one further in the direction of stress took that rating or higher.



9. Unemployment

Level 4.2%, rated 1; change over a year −0.2 points, rated 2. The rating table shows the rate would need to reach 5.4% to rate 2 on its level, and 9.0% to rate 5.

Unemployment rate

BLS (household survey). percent, monthly. Pulled 2026-10-16.

Back to Ratings

Level

1 Low **4.2** 2026-09-01

Sep 2026: 4.2 percent: at or above 23.3% of the 944 readings since Jan 1948. Rated 1; 2 starts at 50%. Rated 1 since Aug 2021 (61 readings).

Change over 12 months

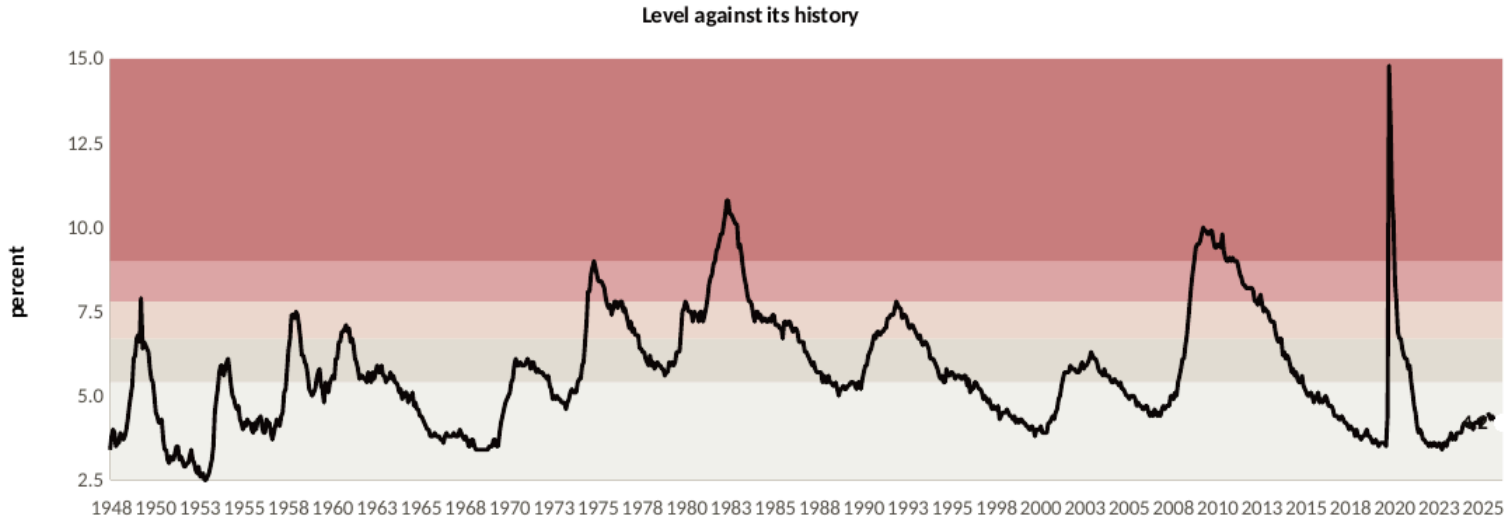
2 Moderate-Low **-0.2** 2026-09-01

Change over 12 months to Sep 2026: -0.2 points: at or above 56.5% of the 932 such changes since Jan 1949. Rated 2; 3 starts at 75%. Rated 2 since Jun 2025 (15 readings).

Where each rating starts

Rating	Level from	Readings	Change from	Changes
1 Low	below 2	438	below 2	405
2 Moderate-Low	5.4	267	-0.3	287
3 Moderate	6.7	135	0.4	143
4 Moderate-High	7.8	50	1.7	49
5 High	9.0	54	2.7	48

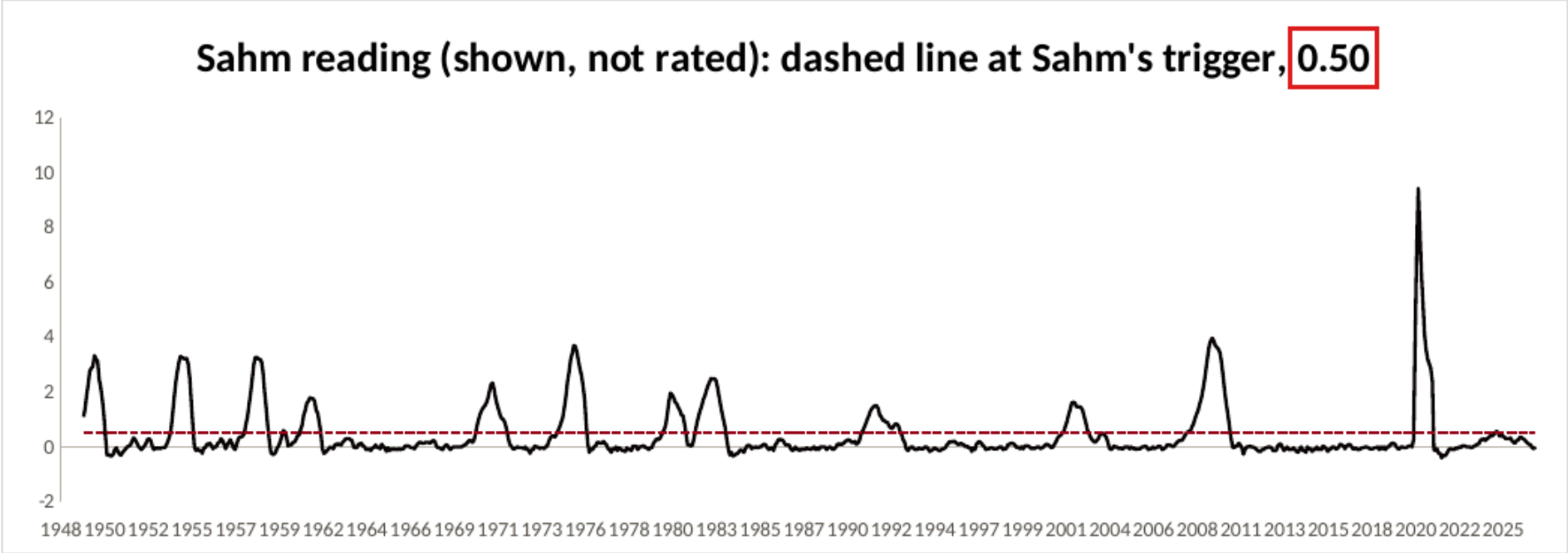
From: the least extreme reading (or change) in this history that took the rating; every one further in the direction of stress took that rating or higher.



Unemployment: both ratings and where each starts

10. The Sahm reading

Further down the unemployment tab: the Sahm reading over its history, with Sahm's published trigger, 0.50, as a dashed line. On 16 October 2026 it is 0.00. It is shown beside the ratings and never used to make them.



What it is

Source	FRED UNRATE
Licence	FRED: Public Domain: Citation Requested
First reading	1948-01
How it is read	Level and change
Which way is worse	higher is worse (from the research)
Break	2025-10: no value: the 2025 lapse in appropriations (BLS footnote 9); kept as missing, never filled
Break	each January: population controls

The Sahm reading and its 0.50 trigger

11. States

States rates each state's unemployment rate the same way, worst first. State data runs a month behind the national rate (August against September on 16 October 2026).

Unemployment by state

BLS (LAUS), pulled 2026-10-16. Each state against its own history, worst first.

State	Latest	Unemployment rate	Level	Level percentile	Change over 12 months	Change	Change percentile	History from
AL	2026-08	3.4	1 Low	12.2%	+0.6	3 Moderate	81.8%	1976-01
AZ	2026-08	4.9	1 Low	34.1%	+0.5	3 Moderate	78.2%	1976-01
CT	2026-08	5.1	2 Moderate-Low	52.1%	+1.0	3 Moderate	83.5%	1976-01
FL	2026-08	4.5	1 Low	27.8%	+0.5	3 Moderate	78.2%	1976-01
HI	2026-08	2.7	1 Low	18.1%	+0.6	3 Moderate	86.2%	1976-01
IL	2026-08	4.7	1 Low	18.3%	+0.4	3 Moderate	76.6%	1976-01
MN	2026-08	4.4	2 Moderate-Low	50.7%	+0.5	3 Moderate	79.7%	1976-01
NM	2026-08	4.7	1 Low	13.8%	+0.6	3 Moderate	82.2%	1976-01
OK	2026-08	4.3	1 Low	45.1%	+0.9	3 Moderate	86.6%	1976-01
VA	2026-08	3.6	1 Low	30.0%	+0.3	3 Moderate	77.6%	1976-01
WA	2026-08	4.9	1 Low	24.4%	+0.3	3 Moderate	75.5%	1976-01
CA	2026-08	5.1	1 Low	18.3%	-0.4	2 Moderate-Low	51.4%	1976-01
CO	2026-08	4.0	1 Low	30.5%	+0.1	2 Moderate-Low	62.2%	1976-01
DE	2026-08	4.7	2 Moderate-Low	50.6%	-0.3	1 Low	47.6%	1976-01
GA	2026-08	3.2	1 Low	4.3%	-0.2	2 Moderate-Low	55.8%	1976-01
ID	2026-08	3.6	1 Low	20.1%	0.0	2 Moderate-Low	63.5%	1976-01
KS	2026-08	3.8	1 Low	27.8%	+0.1	2 Moderate-Low	66.4%	1976-01
KY	2026-08	4.7	1 Low	22.1%	+0.2	2 Moderate-Low	69.6%	1976-01
LA	2026-08	4.2	1 Low	10.4%	0.0	2 Moderate-Low	55.1%	1976-01
MA	2026-08	4.3	1 Low	33.9%	-0.2	2 Moderate-Low	58.2%	1976-01
MD	2026-08	4.1	1 Low	29.0%	-0.2	2 Moderate-Low	52.9%	1976-01

Unemployment by state, worst first

12. Look up a word

Glossary explains every term in the workbook and how it is worked out, from the settings the run used: the cut-offs it names are the ones in force.

Glossary

Every term in this workbook, and how it is worked out.

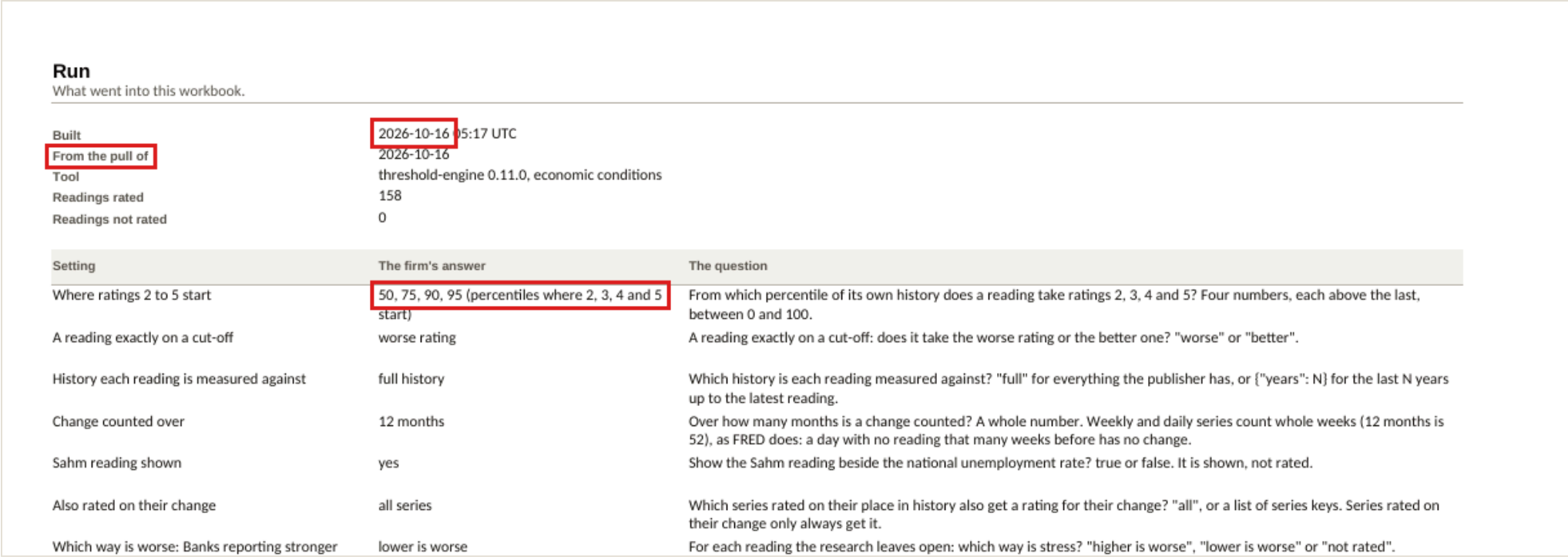
Term	What it means, and how it is worked out
Level	The latest reading itself, compared with every reading in the series' own history.
Change over 12 months	The latest reading minus the reading 12 months earlier. For a rate, a spread or an index around zero it is in points; for a level that grows (prices, jobs, output, credit, claims) it is a percent of the earlier reading. Weekly and daily series count whole weeks (12 months is 52), as FRED does; a day with no reading that many weeks before has no change.
Percentile	The share of the series' history that the latest reading is at or above (where higher is worse) or at or below (where lower is worse). The latest reading counts in its own history, so a new record is 100%.
Rating	1 Low to 5 High, from the percentile: 2 starts at the 50th, 3 at the 75th, 4 at the 90th and 5 at the 95th. A reading exactly on a cut-off takes the worse rating.
From	Where a rating starts, in the series' own units: the least extreme reading in this history that took the rating.
Readings, Changes	How many readings (or changes) in the history took each rating.
History from	The first reading the rating is measured against. Every series is measured against its full history.
At this rating since	The first of the unbroken run of readings, up to the latest, that share the latest rating, each rated against the same history.
What stands out	The sentence behind the higher of the series' two ratings (the change on a tie).
Change only	A level that climbs over the decades with prices and the economy sits near the top of its own history in almost every year, so its place there says nothing. Only its change is rated.
Higher is worse, lower is worse	Which way stress runs for a reading: from the research where it says, otherwise set by the firm. Each series' tab says which.
Sahm reading	The 3-month average unemployment rate minus its lowest 3-month average in the 12 months before. Sahm's published trigger is 0.50. Shown, not rated. Where a month has no value (October 2025), the average is taken from the two months that do, as FRED does.
Pulled	The day the data was downloaded. Every pull is kept in its own folder, so a rating can always be traced to the release it used.
No value published	A period the publisher left blank, such as October 2025 (the lapse in appropriations). It stays empty: never filled, never read as zero.
Made from two series	House prices against income, and the mortgage rate over the 10-year Treasury, are worked out from two series in the same pull. Each tab says how.
Break	A change in how the publisher builds the series, or a known seasonal step, that a reader should know before comparing across it.
Not rated	A reading the settings call for that has no rating: not in the pull, or chosen by the firm not to be rated. The Not rated tab says which and why, when there are any.

The glossary

13. Check what went in

Run says when the workbook was built, from which pull, and every answer in the firm's settings.

Check: *From the pull of* is the folder you rated.



Run

What went into this workbook.

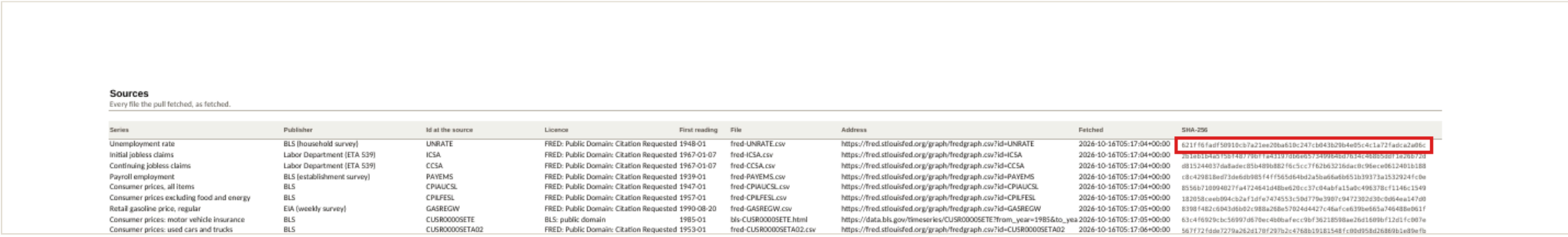
Built	2026-10-16 05:17 UTC
From the pull of	2026-10-16
Tool	threshold-engine 0.11.0, economic conditions
Readings rated	158
Readings not rated	0

Setting	The firm's answer	The question
Where ratings 2 to 5 start	50, 75, 90, 95 (percentiles where 2, 3, 4 and 5 start)	From which percentile of its own history does a reading take ratings 2, 3, 4 and 5? Four numbers, each above the last, between 0 and 100.
A reading exactly on a cut-off	worse rating	A reading exactly on a cut-off: does it take the worse rating or the better one? "worse" or "better".
History each reading is measured against	full history	Which history is each reading measured against? "full" for everything the publisher has, or {"years": N} for the last N years up to the latest reading.
Change counted over	12 months	Over how many months is a change counted? A whole number. Weekly and daily series count whole weeks (12 months is 52), as FRED does: a day with no reading that many weeks before has no change.
Sahm reading shown	yes	Show the Sahm reading beside the national unemployment rate? true or false. It is shown, not rated.
Also rated on their change	all series	Which series rated on their place in history also get a rating for their change? "all", or a list of series keys. Series rated on their change only always get it.
Which way is worse: Banks reporting stronger	lower is worse	For each reading the research leaves open: which way is stress? "higher is worse", "lower is worse" or "not rated".

The Run tab: the pull and the settings

14. Check a file

Sources lists every file the pull fetched: its address, licence, when it was fetched and its SHA-256 fingerprint. The fingerprint is the same as in the pull folder's `manifest.json`; a different one means the workbook was made from different data.



Sources

Every file the pull fetched, as fetched.

Series	Publisher	Id at the source	Licence	First reading	File	Address	Fetched	SHA-256
Unemployment rate	BLS (household survey)	UNRATE	FRED: Public Domain: Citation Requested	1948-01	fred-UNRATE.csv	https://fred.stlouisfed.org/graph/fredgraph.csv?id=UNRATE	2026-10-16T05:17:04+00:00	621ff6fadf50910cb7a21ee20ba610c247cb043b29b4e05c4c1a72fadca2a06c
Initial jobless claims	Labor Department (ETA 539)	ICSA	FRED: Public Domain: Citation Requested	1967-01-07	fred-ICSA.csv	https://fred.stlouisfed.org/graph/fredgraph.csv?id=ICSA	2026-10-16T05:17:04+00:00	2b1ab1b4a515bf4b799bff443197d6be6572499b4bd7b34c488b5d0f1e26b72d
Continuing jobless claims	Labor Department (ETA 539)	CCSA	FRED: Public Domain: Citation Requested	1967-01-07	fred-CCSA.csv	https://fred.stlouisfed.org/graph/fredgraph.csv?id=CCSA	2026-10-16T05:17:04+00:00	d8152440370a8adec85b489bb82f6c5cc7f62b63216dac0c96ece0612401b188
Payroll employment	BLS (establishment survey)	PAYEMS	FRED: Public Domain: Citation Requested	1939-01	fred-PAYEMS.csv	https://fred.stlouisfed.org/graph/fredgraph.csv?id=PAYEMS	2026-10-16T05:17:04+00:00	c8c429818ed73de6db985f4ff565d64bd2a5ba66a6b651b39373a1532924fc0e
Consumer prices, all items	BLS	CPIAUCSL	FRED: Public Domain: Citation Requested	1947-01	fred-CPIAUCSL.csv	https://fred.stlouisfed.org/graph/fredgraph.csv?id=CPIAUCSL	2026-10-16T05:17:04+00:00	8556b710094027fa4724641d48be620cc37c04abfa15a0c496378cf1146c1549
Consumer prices excluding food and energy	BLS	CPILFESL	FRED: Public Domain: Citation Requested	1957-01	fred-CPILFESL.csv	https://fred.stlouisfed.org/graph/fredgraph.csv?id=CPILFESL	2026-10-16T05:17:05+00:00	182058ceeb094cb2af1dfe7474553c50d779e3907cf9472302d30c0d64ea147d0
Retail gasoline price, regular	EIA (weekly survey)	GASREGW	FRED: Public Domain: Citation Requested	1990-08-20	fred-GASREGW.csv	https://fred.stlouisfed.org/graph/fredgraph.csv?id=GASREGW	2026-10-16T05:17:05+00:00	8398f482c6043d6b02c988a268e57024d4427c46afce639be665a74648be061f
Consumer prices: motor vehicle insurance	BLS	CUSR0000SETE	BLS: public domain	1985-01	bls-CUSR0000SETE.html	https://data.bls.gov/timeseries/CUSR0000SETE?from_year=1985&to_yea	2026-10-16T05:17:05+00:00	63c4f6929cbc56997d670ec4b6bafecc9bf36218598ae26d1609bf12d1fc007e
Consumer prices: used cars and trucks	BLS	CUSR0000SETA02	FRED: Public Domain: Citation Requested	1953-01	fred-CUSR0000SETA02.csv	https://fred.stlouisfed.org/graph/fredgraph.csv?id=CUSR0000SETA02	2026-10-16T05:17:06+00:00	567f72fdde7279a262d170f297b2c4768b19181548fc00d958d26869b1e89efb

Sources: the unemployment file's fingerprint



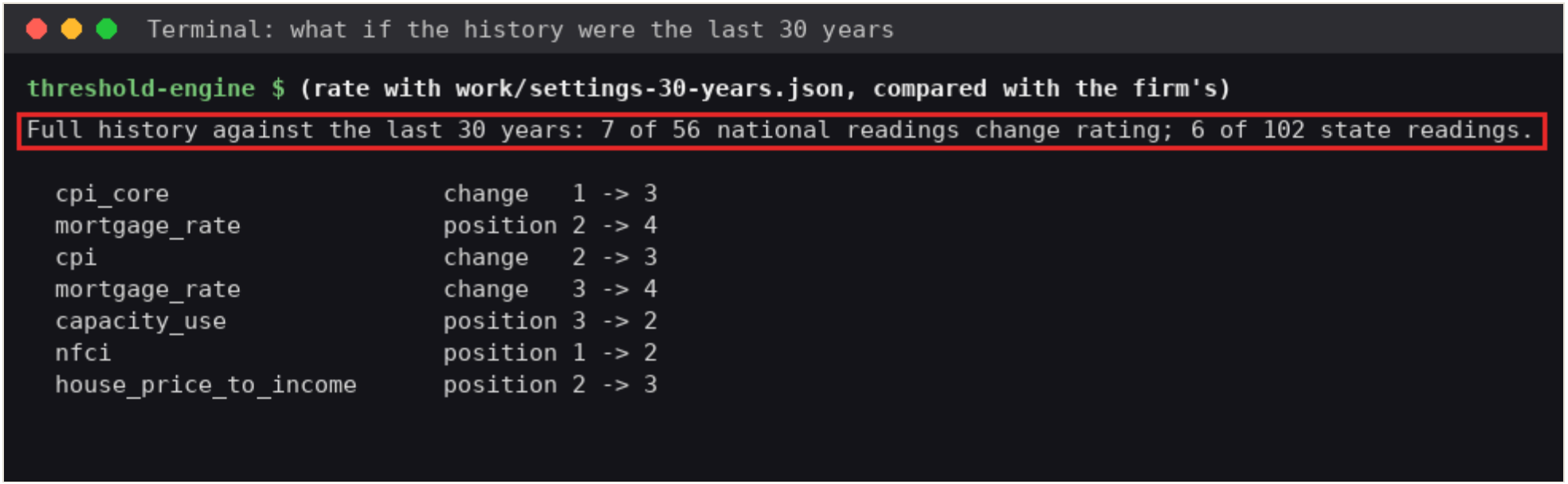
The fingerprint, close up

Part 3: what if, and what refuses

15. Try another setting without changing the firm's

Copy `docs/macro/settings.json` to `work/`, change an answer in the copy (here the history, from `"full"` to `{"years": 30}`), and rate with `--settings` pointing at the copy. Only answers are read.

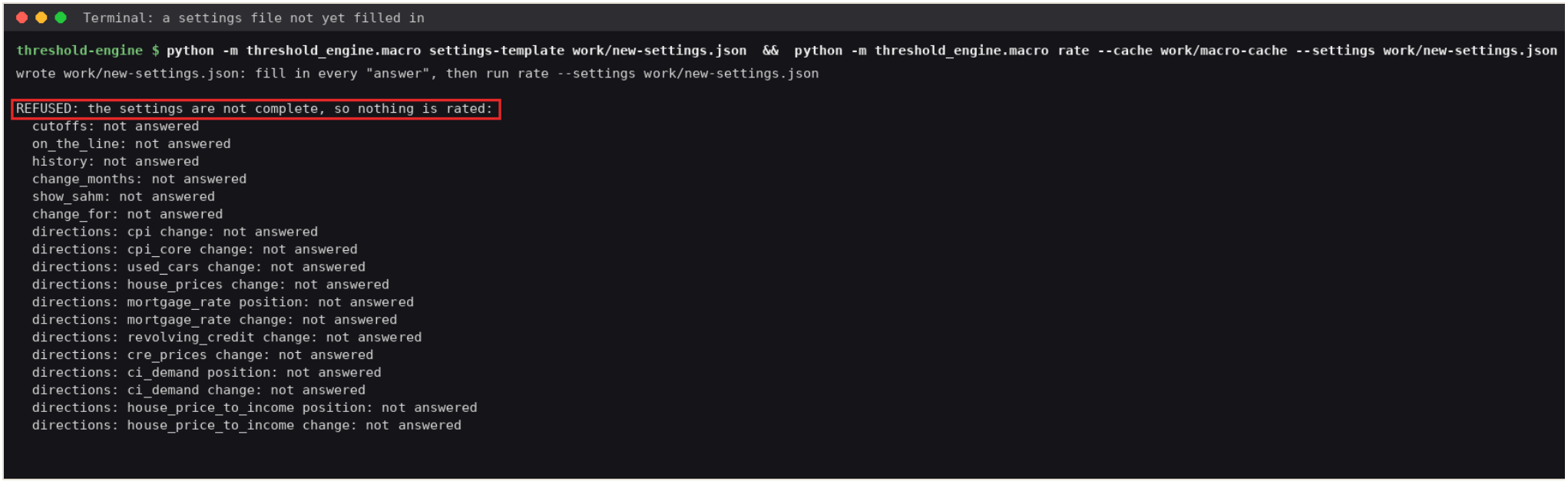
Against the last 30 years rather than full history, 7 of the 56 national readings change rating; the mortgage rate's level goes from 2 to 4. The window matters, as the research warned. (Comparing the two runs took a short script: see the walk's defects, W12.)



The readings whose rating changes over 30 years

16. A settings file not yet filled in

`settings-template` writes the questions with every answer empty. Rate with it and the tool refuses, naming every gap at once, and rates nothing.



Refused, with every missing answer named

When you are done

The pull and the workbook are in `work/` inside the repository. Nothing tells git to leave `work/` alone yet (defect W1), so delete it or move it before committing anything.

A wrong screen

- `pulled 84 of 85` or a `FAILED` line: a publisher did not answer. Pull again later; the earlier folder is kept either way.
- `REFUSED:` from `rate`: the settings file is missing an answer. The message names each one.
- A Ratings row with no rating: open **Not rated**, which says why.
- A chart line that drops to zero: it should never happen. A month with no value is an empty cell and a gap in the line. Report it.